

Nithish Krishnabharathi GNANI

Polytech Nice Sophia, France | [linkedin.com/in/nithish-k-gnani/](https://www.linkedin.com/in/nithish-k-gnani/) | nkgnani101@gmail.com

EDUCATION

- Polytech Nice Sophia, Université Cote d'Azur
- KTH Royal Institute of Technology

EIT Digital Dual Master of Science - Autonomous Systems and Intelligent Robots
Minor: Innovation and Entrepreneurship

Sophia Antipolis, France
Stockholm, Sweden
2024 - 2026

- National Institute of Technology Karnataka (NITK) | top 10 engineering institution in India
Bachelor of Technology (4-year program) - Mechanical Engineering

Surathkal, India
2013 - 2017

WORK EXPERIENCE

Indian Institute of Science
Technical Associate

Bangalore, India
2019-2024

- Led industry-funded research projects. Sponsored by the Cisco Center for Networked Intelligence at Robert Bosch Center for Cyber-Physical Systems (RBCCPS). Promoted from Research Assistant to this role.

Coca-Cola (HCCBPL)
Team Leader, Maintenance

Hyderabad, India
2017-2018

- Managed a team of 20+ associates to achieve consistent reliability of utility systems.

SELECTED PROJECTS (More: <https://nithishkgnani.github.io/projects/>)

BalanceBot: Simulink-Based Control Deployment on Hardware

Sep 2024 - present

- To implement various control algorithms on a two-wheeled robot to move and balance it using a model based approach.

Designing Tactile Cyber-Physical Systems

Mar 2022 - Aug 2024

- Created a testbed for real-time human-robot interaction with ultra-reliable low latency communication. Built a remote robotic arm system interfaced with a haptic device for teleoperation across continents.
- Implemented IEEE 802.1 TSN features on hardware for enhanced network reliability and latency.

5G AMMAZING

Mar 2020 - Jun 2022

- Developed a 5G mmWave system for infotainment with split data and control planes over 5G and Wi-Fi.
- Won first place (€6,500 cash award) in 5G Hackathon 2020 by Government of India among 1100+ teams.

Wi-Fi Modelling

Jul 2020 - Jun 2021

- Modelling & simulation of IEEE 802.11n & 802.11ac using MATLAB to evaluate throughput, packet error rate & range.

Acoustic PUF for Sensor Security

Nov 2019 - Jul 2021

- Identified and localized sensor devices using acoustic fingerprinting for security and integrity monitoring.

Energy Harvesting IoT Sensors

Nov 2019 - Jun 2020

- Developed adaptive sampling algorithms for energy-efficient data management in IoT sensors.

Total Productive Maintenance

Nov 2017 - Apr 2018

- Improved reliability and reduced failures in Coca-Cola's bottling plant through TPM steps.

SKILLS & INTERESTS

Programming : Python, MATLAB, Simulink, P4, C, C++, Embedded C, Shell
Hardware : Netronome Agilio Smart NIC, Nordic nRF52 series, Arduino
Tools : Linux, Git, ROS, Docker, LaTeX, Autodesk Fusion 360
Creative : 3D printing, Shotcut, Snapseed
Languages : English (IELTS band score 8/9), French (A1)



Website

PUBLICATIONS

- EdgeP4: In-Network Edge Intelligence for a Tactile Cyber-Physical System Testbed Across Cities
- Towards a TSN-DetNet Intercity Testbed for Tactile Cyber-Physical Systems
- Sensor Identification via Acoustic Physically Unclonable Function
- Judicious data management for sustaining an energy harvesting sensor node

Best Paper Award
IEEE INFOCOM '24
IEEE INFOCOM '24
ACM DTRAP journal
Wiley CCPE journal